



DPUVidCapture

You can use DPUVidCap to create digital videos during the simulation. The videos contain the image of an external camera.

The file format of the recorded video files is WindowsAVI (.avi).

Any installed DirectShow video codec can be used to compress the video.

Good results can be achieved with the DivX and the x264vfw codecs.

1. Parameters for naming the files:

<i>Path</i>	Output path of the recorded file. If no absolute path is provided, the path is located below SILAB's installation path. Default: "VIDEO\"
<i>FileName</i>	File name of the created video (within the <i>Path</i> folder). Default: "Video" Alternatively, it can be specified using the launch data <i>VAR_DATA-FILENAME</i> .
<i>Suffix</i>	<i>Suffix</i> that is inserted into the file name (after <i>FileName</i> , but before the extension ".avi").
<i>OverwriteExistingFile</i>	If set to <i>true</i> , an already existing video file is overwritten. If set to <i>false</i> (the default), an error message is issued and the simulation cannot be started if the output file already exists.
<i>LD_SaveDataToken</i>	The name of a LaunchData entry (of type LD_BOOL), which defines if a video file is created. Default: <i>VAR_SAVEDATA</i>
<i>LD_DataFileNameToken</i>	The name of a LaunchData entry (of type LD_EDIT), which defines the video file name (instead of the <i>FileName</i> parameter). Default: <i>VAR_DATAFILENAME</i>

2. Parameters concerning video capture:

<i>VideoSource</i>	The name of the external video source that is used for the capture. (The default is an empty string, which means the first available video source is used.) Special feature: In order to capture video and/or audio data from a DV camera, <i>VideoSource</i> and <i>AudioSource</i> must be set to "DV".
<i>AudioSource</i>	The name of an audio device that is used for the capture. (The default is an empty string, which means the first available audio source is used.)
<i>VideoWidth, VideoHeight</i>	Desired width and height of the video captured from the camera. If not specified, a camera-dependent default value is used. Note: Most cameras support only a few fixed image formats. If the specified <i>VideoWidth</i> is not available, the nearest supported format



	is selected. Some formats do not function reliably even though they are advised by the camera.						
<i>VideoFPS</i>	Desired frame rate (frames per second) of the captured video. If not specified, a camera-dependent default value is selected. Note: Most cameras support only a few fixed frame rates. If the specified FPS is not available, the nearest supported frame rate is selected. Some rates do not function reliably even though they are advised by the camera. In addition to <i>VideoFPS</i>, <i>VideoWidth</i> and <i>VideoHeight</i> must be defined to select a frame rate.						
<i>AudioLatencyMS</i>	Latency of the audio capture in ms. (Default: 200) Lower values can lead to stuttering of the sound, values > 250 may result in worse audio/video synchronization.						
<i>SyncReactSpeed</i>	Gibt an, wie schnell die Audio-Video-Synchronisation auf Änderungen in der Video-Bildrate reagieren soll. Gültige Werte sind 0.1 bis 0.9. Der Standardwert 0.5 ist ein guter Kompromiss zwischen besserer Synchronisation und allzu starken Tonhöhenschwankungen.						
<i>AudioOffsetMS</i>	Can be used to correct a sync error in the created video file. This can especially happen if separate cards are used for audio and video capture (e.g. a discrete video capture card and the PC's built-in sound card).						
<i>InterleaveMethod</i>	Positive values play audio earlier, negative values play it later. Synchronization method for audio and video data in the created AVI file. <table><tr><td>0 (default)</td><td>No synchronization. (Usually works well because DPUVidCap performs the synchronization instead.)</td></tr><tr><td>1</td><td>„Capture“ synchronization. (Doesn't work well, usually.)</td></tr><tr><td>2</td><td>Complete synchronization. (Suitable for cameras with integrated microphones like webcams and DV cameras.)</td></tr></table>	0 (default)	No synchronization. (Usually works well because DPUVidCap performs the synchronization instead.)	1	„Capture“ synchronization. (Doesn't work well, usually.)	2	Complete synchronization. (Suitable for cameras with integrated microphones like webcams and DV cameras.)
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3. Parameters concerning video compression:

<i>VideoCodec</i>	Name of the video codec that is used for compression. Default value: "DivX". If the name is empty (<i>VideoCodec</i> = ""), uncompressed video will be created (this is not recommended).
<i>Quality</i>	Quality setting for the video codec (range: 0..100, default: not set). Most video codecs ignore the quality setting.
<i>KeyFrameRate</i>	Specifies the number of key frames in the created video. Values around 30 are recommended (i.e. about one key frame per second). If not specified, the video codec's default setting is used.
<i>CompressionOptions</i>	File which contains additional compression options for the video codec. If the file does not exist, a dialog is opened in which the options can be configured. They are then saved into this file. If a relative path is specified, the file is searched / created below SILAB's installation path.



4. Parameters for the video preview:

<i>Preview</i>	If set to true, a preview window is shown which displays the output video. If set to false (the default), the video is only written to the file.
<i>PreviewX, PreviewY</i>	Position of the preview window on screen (in pixels).

5. Outputs:

<i>VideoTimeMS</i>	Current video time stamp, in milliseconds. If a DPUDataFile recording is running in parallel, it is recommended to add this variable to the DPUDataFile, so that the files can be synchronized precisely afterwards.
<i>Out_AudioSpeed</i>	Speed of the audio source in relation to the video source.
<i>Out_AudioDelayMS</i>	Current delay of the audio source vs. the video source.
<i>Out_AudioScale</i>	Current stretch value for the audio data.
<i>Out_NDroppedVideo</i>	Counts the number of video frames that had to be dropped during the whole simulation due to CPU overload. This value should always be less than five; otherwise some parameters must be reduced.
<i>Out_NDroppedAudio</i>	Counts the number of audio samples that had to be dropped during the whole simulation.

6. Recommendations:

- Encoding videos in real-time imposes high demands on the computer. Depending on the video resolution, a current dual or quad core processor is recommended.

7. Functional principle:

The DPU works well when using the following script:

```
DPUVidCap vidcap
{
    Computer = {DATA};
    Index = 100;

    VideoCodec = "DivX 6";
    CompressionOptions = "lib\\configurations\\VidCap\\DivX6_PAL.vcc";

    #AudioSource = Realtek;

    AudioOffsetMS = 0;

    SyncReactSpeed = 0.5;
    InterleaveMethod = 0;

    Preview = true;
    PreviewX = 10;
    PreviewY = 10;
```



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```
VideoWidth = 640;
VideoHeight = 480;
VideoFPS = 25;

SaveFile = true;
OverwriteExistingFile = true;
Path = "\\Videos\\";
FileName = "VidCap";
};

DPUDataFile data
{
    Computer = {DATA};
    Index = 110;
    Path = "\\Videos\\";
    FileName = "VidCap";
    HeaderType = 1;
    OverwriteExistingFile = true;

    Record = {
        vidcap.VideoTimeMS
    };
};
```